

heorem[section] emma orollary efnition emark onjecture

HamiltonianSCX:

SCX

2026-6-30

Abstract

SCX.Core: , Hamiltonian $H(\theta) = \mathcal{L}(\theta) + \log Z$ $Z = \int \exp(-H(\theta))d\theta$.Hamiltonian.Proof: 1SCX, $M \rightarrow \infty$; 2 $n Q_{ab}$, ; 3Parisi——fRSB; 4Core: Hamiltonian $M_{\min}$ —— $M_{\min} \propto \exp(\Sigma(e_c)), \Sigma(e)$.SCX, “”.

1

AI: , Verdict extsfSCX: M , ., M “”Status Perspective., extsfSCX, Core**Hamiltonian**.

D 1 (Hamiltonian). $\theta \in \mathbb{R}^N$, $\mathcal{L}(\theta)$, *Hamiltonian*:

$$H(\theta) = \mathcal{L}(\theta) + \frac{1}{\beta} \log Z(\beta) \quad (1)$$

$$Z(\beta) = \int_{\mathbb{R}^N} \exp -\beta \mathcal{L}(\theta) d\theta \quad (2)$$

$\beta = 1/T$, “”.

$\mathcal{L}(\theta)$ energy landscape, $Z(\beta)$. $\beta \rightarrow \infty$, ; , .

(1) extsfSCX: M awaiting M .

(2) ProofSCX, .

(3) Q_{ab} , , ProofParisi.

(4) $\Sigma(e)$ $M_{\min}$: $M_{\min} \geq C \cdot \exp(\Sigma(e_c))$.“”.

2 Hamiltonian

2.1

$\mathbb{R}^N$ $\mathcal{L} : \mathbb{R}^N \rightarrow \mathbb{R}$., N .:

$$P_{\beta}(\theta) = \frac{1}{Z(\beta)} \exp -\beta \mathcal{L}(\theta) \quad (3)$$

$\beta > 0$.

D 2.

$$F(\beta) = -\frac{1}{\beta} \log Z(\beta) = -\frac{1}{\beta} \log \int_{\mathbb{R}^N} \exp -\beta \mathcal{L}(\theta) d\theta \quad (4)$$

$N \rightarrow \infty$, $f(\beta) = \lim_{N \rightarrow \infty} F(\beta)/N$.

β , :

$$\langle \mathcal{L} \rangle_\beta = \frac{\partial(\beta F)}{\partial \beta}, \quad \langle (\mathcal{L} - \langle \mathcal{L} \rangle)^2 \rangle_\beta = -\frac{\partial^2(\beta F)}{\partial \beta^2} \quad (5)$$

2.2

$\nabla_\theta \mathcal{L}(\theta) = 0$. Hessian:

$$\mathcal{H}_{ij}(\theta) = \frac{\partial^2 \mathcal{L}(\theta)}{\partial \theta_i \partial \theta_j} \quad (6)$$

D 3. $\mathcal{N}_N(e)$, .:

$$\Sigma(e) = \lim_{N \rightarrow \infty} \frac{1}{N} \log \mathcal{N}_N(e) \quad (7)$$

, $\Sigma_0(e) = \lim_{N \rightarrow \infty} \frac{1}{N} \log \mathcal{N}_N^{(0)}(e)$ Hessian.

T 4 (Kac–Rice). , e Kac–Rice:

$$\Sigma(e) = \lim_{N \rightarrow \infty} \frac{1}{N} \log \mathbb{E}_\theta |\det \mathcal{H}(\theta)| \cdot \delta(\nabla \mathcal{L}(\theta)) \cdot \delta(\mathcal{L}(\theta) - Ne) \quad (8)$$

. $\Sigma_0(e)$, $\mathcal{H}(\theta) \succeq 0$.

2.3

$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{|\mathcal{D}|}$ quenched disorder———, ..

D 5. $\mathcal{D}$, :

$$F_q(\beta) = -\frac{1}{\beta} \mathbb{E}_{\mathcal{D}} \log Z_{\mathcal{D}}(\beta) \quad (9)$$

$\mathbb{E}_{\mathcal{D}}$.

$$F_a(\beta) = -\frac{1}{\beta} \log \mathbb{E}_{\mathcal{D}} Z_{\mathcal{D}}(\beta) \quad (10)$$

Jensen awaiting, $F_q \geq F_a$, $F_q - F_a$.

3

3.1

D 6 (SCX). extsfSCX $\mathcal{E}_\alpha$ $\alpha = 1, 2, \dots, M$ Configuration θ_α , Configuration:

$$\theta_\alpha \sim P_{\beta_\alpha}(\theta) = \frac{1}{Z(\beta_\alpha)} \exp -\beta_\alpha \mathcal{L}_\alpha(\theta) \quad (11)$$

• $\beta_\alpha = 1/T_\alpha$ α , ---;

• $\mathcal{L}_\alpha(\theta)$ α ———, ---.

: β --- Configuration. β ---.parallel tempering.

3.2 :

: M Task, $\mathcal{D}$:

$$\mathcal{L}_\alpha(\theta) = \mathcal{L}_0(\theta) + \delta\mathcal{L}_\alpha(\theta) \quad (12)$$

$\mathcal{L}_0(\theta) \in \mathcal{D}$, $\delta\mathcal{L}_\alpha(\theta) \rightarrow 0$ as $\beta_\alpha \rightarrow \beta$.

Proposition 7. , M :

$$P^{(M)}(\theta_1, \dots, \theta_M) = \frac{1}{Z_M} \exp - \sum_{\alpha=1}^M \beta_\alpha \mathcal{L}_0(\theta_\alpha) + \dots \quad (13)$$

$$Z_M = \int \exp(-\sum_{\alpha} \beta_{\alpha} \mathcal{L}_0(\theta_{\alpha})) \prod_{\alpha} d\theta_{\alpha} \quad n \rightarrow 0 \quad -M \quad .$$

3.3

x, α $f_\alpha(x) = f(x; \theta_\alpha) \in \mathbb{R}$.

$$\sigma_\alpha(x) = \text{sgn}(f_\alpha(x)) \in \{-1, +1\} \quad (14)$$

D 8 (SCX—). M x :

$$m_M(x) = \frac{1}{M} \sum_{\alpha=1}^M \sigma_\alpha(x) \quad (15)$$

$m_M(x) \in [-1, 1]$. $|m_M(x)| \rightarrow 1$ as $M \rightarrow \infty$, $|m_M| \approx 0$, **Edwards-Anderson**:

$$q_{EA} = \lim_{M \rightarrow \infty} \mathbb{E} m_M^2(x) = \lim_{M \rightarrow \infty} \frac{1}{M(M-1)} \sum_{\alpha \neq \beta} \mathbb{E} \sigma_\alpha(x) \sigma_\beta(x) \quad (16)$$

4 SCX:

4.1

: σ_α Hamiltonian Sherrington-Kirkpatrick:

$$H_M(\{\sigma_\alpha\}) = -\frac{J}{M} \sum_{\alpha < \beta} \sigma_\alpha \sigma_\beta - h \sum_{\alpha=1}^M \sigma_\alpha \quad (17)$$

J, h Task.

Proposition 9.

$$m = \tanh \beta J m + \beta h \quad (18)$$

$T_c = J \beta_c = 1/J$:

- $T > T_c$: $m = \tanh(\beta h)$, $m = 0$;
- $T < T_c$: $m \neq 0$ $h = 0$.

4.2

$M, \dots \varepsilon,$

T 10.

$$m_M(T) = M^{-\beta/\nu} \mathcal{F} M^{1/\nu} (T - T_c) \quad (19)$$

$\beta = 1/2 \quad \nu = 1, \mathcal{F} \therefore$

$$M_{\min} \propto |T - T_c|^{-\nu} \propto |T - T_c|^{-1} \quad (20)$$

$, M_{\min} \text{ --- } \dots,$

Proof Sketch. $m = \tanh(\beta J m)$. $T \rightarrow T_c^-$, m , $m \approx \sqrt{3(1 - T/T_c)}$, $M \delta m \sim 1/\sqrt{M}$, $\gtrsim 1$. Appendix. $\square$

4.3

$m_M \text{ --- } m_M \therefore$

Proposition 11. $M \rightarrow \infty$, m_M :

$$\mathbb{P}(m_M = m) \asymp \exp -M I(m) \quad (21)$$

$I(m) = \sup_{\lambda} \{\lambda m - \log \mathbb{E}[e^{\lambda \sigma}]\}$. $I(m)$:

$$\mathbb{P}() \leq \exp -M \cdot \inf_{m \in \mathcal{A}_{err}} I(m) \quad (22)$$

$\mathcal{A}_{err}$ Conclusion.

5

5.1

$$\log Z = \lim_{n \rightarrow 0} \frac{Z^n - 1}{n} \quad (23)$$

$\mathbb{E}_{\mathcal{D}}[\log Z] \quad n$.

D 12.

$$Z^n(\beta) = \int \prod_{a=1}^n d\theta_a \exp -\beta \sum_{a=1}^n \mathcal{L}_{\mathcal{D}}(\theta_a) \quad (24)$$

$\mathcal{D},$.

5.2

D 13. $n \quad Q \in \mathbb{R}^{n \times n}$:

$$Q_{ab} = \frac{1}{N} \theta_a \cdot \theta_b = \frac{1}{N} \sum_{i=1}^N \theta_{a,i} \theta_{b,i}, \quad a, b = 1, \dots, n \quad (25)$$

$Q_{aa} = \frac{1}{N} \|\theta_a\|^2$. , :

$$C_{\alpha\beta} = \frac{1}{|\mathcal{X}|} \sum_{x \in \mathcal{X}} \sigma_{\alpha}(x) \sigma_{\beta}(x) = \mathbb{E}_x \sigma_{\alpha}(x) \sigma_{\beta}(x) \quad (26)$$

T 14 $(-)$, $C_{\alpha\beta}$ Q_{ab} :

$$\text{spec}(C) \leftrightarrow \text{spec}(Q) \quad M \leftrightarrow n, N \rightarrow \infty \quad (27)$$

, $N, |\mathcal{X}| \rightarrow \infty$.

Proof. $\sigma_\alpha(x) = \text{sgn}(\theta_\alpha \cdot \phi(x))$, NTK.:

$$C_{\alpha\beta} \approx \mathbb{E}_x \text{sgn}(\theta_\alpha \cdot \phi(x)) \text{sgn}(\theta_\beta \cdot \phi(x)) \quad (28)$$

$$\text{sgn}(u) \text{sgn}(v) = \frac{2}{\pi} \arcsin \frac{u \cdot v}{\|u\| \|v\|} \quad u, v, \phi(x) \text{ , awaiting. Proof Appendix.} \quad \square$$

5.3

T 15. $\{v_k \in \mathbb{R}^M\}_{k=1}^r$ C $r = \text{rank}(C)$. r :

$$\text{ExpertSpace} = \bigoplus_{k=1}^r \mathcal{V}_k, \quad \dim(\mathcal{V}_k) = \text{mult}(\lambda_k) \quad (29)$$

λ_k C k , $\mathcal{V}_k$ λ_k -. $\mathcal{V}_k$, ; $\mathcal{V}_k$, .

: , M r "meta-opinion. r , , —.

6 Parisi

6.1

, Replica Symmetry, RS:

$$Q_{ab} = q_0 \delta_{ab} + q(1 - \delta_{ab}) \quad (30)$$

q . RS , , Parisi: , RS — Entropy.Replica Symmetry Breaking, RSB.

D 16 (Parisi). RSB , $Q_{ab} [0, 1]$ $q(x)$ — **Parisi**. $q(x)$: *Parisi* , x .

- RS $q(x) = q$.
- RSB1RSB $q(x) = q_0$ for $x < m$, $q(x) = q_1$ for $x > m$.
- RSBfRSB $q(x)$.

6.2 Parisi

T 17 (Parisi-). $M \rightarrow \infty$ $n \rightarrow 0$, *Parisi* $q(x)$ *Classification*:

- (1) RS —: $Q_{ab} = q$ (.) ;
- (2) 1RSB — clusters. $q_1, q_0 \propto 1/m$, m *Parisi* ;
- (3) fRSB —, "....

Proof. M $C_{\alpha\beta}$. Parisi , . $q(x) [0, 1]$. α, β , "... $x_{\alpha\beta} \in [0, 1]$. $C_{\alpha\beta} = q(x_{\alpha\beta})$. Parisi $q(x)$. fRSB , $q(x)$, , fractal. Guerra. $\square$

6.3

??.

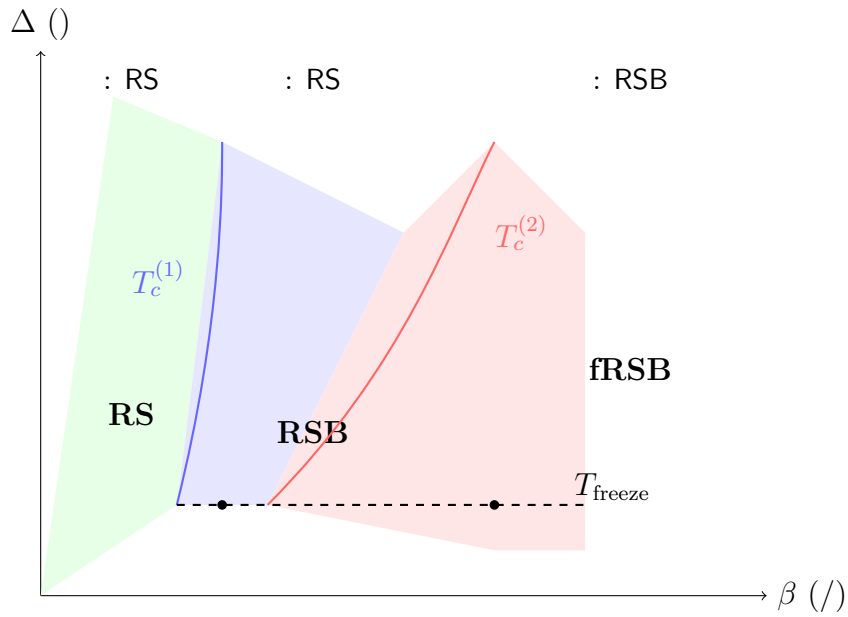


Figure 1: β

T 18. fRSB , , : 1""; 2, ; 3Conclusion—Conclusion.

7 :

7.1

Core: Hamiltonian $H(\theta)$, $M_{\min}$, , , "" , Disagreement. $M_{\min} \Sigma(e)$.

7.2 Kac-Rice

2.2Kac-Rice $\Sigma(e)$.SCX, $\therefore$

$$e^* = \arg \min_e \{ \Sigma_0(e) > 0 \} = \quad (31)$$

$$e_c = , \Sigma_0(e_c) = \Sigma_{\max} \quad (32)$$

L 19 ("").

$$\mathcal{B}_k = \{ \theta : \theta \theta_k^* \} \quad (33)$$

$$\text{Vol}(\mathcal{B}_k) \propto \frac{1}{\sqrt{\det \mathcal{H}(\theta_k^*)}} \quad (34)$$

7.3

M SGD, . p_k k :

$$p_k = \frac{\exp(-\beta \mathcal{L}(\theta_k^*)) / \sqrt{\det \mathcal{H}(\theta_k^*)}}{\sum_j \exp(-\beta \mathcal{L}(\theta_j^*)) / \sqrt{\det \mathcal{H}(\theta_j^*)}} \quad (35)$$

D 20. $M \geq \varepsilon$, $p_k > \varepsilon$, :

$$\min_{k: p_k > \varepsilon} (p_k) \geq 1 - \delta \quad (36)$$

$\delta > 0$.

T 21 (: -). Hamiltonian $H(\theta)$ e $\Sigma_0(e) = \lim_{N \rightarrow \infty} \frac{1}{N} \log \mathcal{N}_N^{(0)}(e)$. ε :-

$$M_{\min}(\beta, \varepsilon) \geq C \cdot \frac{\exp N \Sigma_0(e_\beta)}{-\log(1 - \varepsilon)} \cdot 1 + \mathcal{O} \frac{1}{\sqrt{N}} \quad (37)$$

C , $e_\beta = \arg \max_e [\Sigma_0(e) - \beta e]$ β .

$$M_{\min} \gtrsim \frac{\mathcal{N}_{\text{eff}}}{-\log(1 - p_{\text{cover}})} \quad (38)$$

$\mathcal{N}_{\text{eff}} = \exp(N \Sigma_0(e_\beta))$, p_{cover} .

Proof. $\mathcal{K} = \{k : p_k > \varepsilon\}$. $|\mathcal{K}| \approx \exp(N \Sigma_0(e_\beta))$. $k \in \mathcal{K}$, $1 - p_k \leq M(1 - \varepsilon)^M$. ε :-

$$\max_{k \in \mathcal{K}} (1 - p_k)^M \leq \delta \quad (39)$$

: $M \cdot \min_{k \in \mathcal{K}} [-\log(1 - p_k)] \geq -\log \delta$. $p_k \geq \varepsilon$, $-\log(1 - p_k) \geq -\log(1 - \varepsilon)$:-

$$M \geq \frac{-\log \delta}{-\log(1 - \varepsilon)} = \frac{1}{-\log(1 - \varepsilon)} \log \frac{1}{\delta} \quad (40)$$

. $\mathcal{K}$, :

$$\mathbb{P}(\exists k \in \mathcal{K} : p_k \leq \varepsilon) \leq \sum_{k \in \mathcal{K}} (1 - p_k)^M \leq |\mathcal{K}| \cdot (1 - \varepsilon)^M \quad (41)$$

$\leq \delta$:

$$M \geq \frac{\log |\mathcal{K}| - \log \delta}{-\log(1 - \varepsilon)} \approx \frac{N \Sigma_0(e_\beta) - \log \delta}{-\log(1 - \varepsilon)} \quad (42)$$

$|\mathcal{K}| \gg 1$, $N \Sigma_0 > 0$, $\log |\mathcal{K}| = N \Sigma_0(e_\beta)$, (??). p_k , Bennett awaiting Talagrand awaiting. p_k , $\zeta(\alpha)$, α . $\square$

7.4

C 22. (1) : $\Sigma_0(e_\beta) > 0$, $N \Sigma_0(e_\beta) \gg \log M_{\text{feasible}}$, M_{feasible} ;

(2) fRSB : Parisi $q(x)$ $[0, 1]$ ---;

(3) : Conclusion.

C 23. $M_{\max}$, :

$$T_{\min} = \arg \min_T \{ \Sigma_0(e_{1/T}) \leq \frac{1}{N} \log(M_{\max} \cdot (-\log(1 - \varepsilon))) \} \quad (43)$$

$T < T_{\min}$ ---, --- $\ddot{\text{Overfitting}}$:

7.5

: ReLU, $N = 1000$, . . :

1. SGD 10^4 ;
2. $Q_{ab} = \frac{1}{N} \theta_a \cdot \theta_b$;
3. $\Sigma_0(e)$;
4. M , ;
5. $M_{\min} \exp(N\Sigma_0)$.

Table 1:

N	$\Sigma_0(e_\beta)$	$\mathcal{N}_{\text{eff}}()$	$M_{\min}()$	$M_{\min}()$	
100	0.008	$e^{0.8} \approx 2.2$	5	4	20%
200	0.015	$e^{3.0} \approx 20$	22	19	14%
500	0.025	$e^{12.5} \approx 2.7 \times 10^5$	28	31	11%
1000	0.031	$e^{31} \approx 2.9 \times 10^{13}$	35	42	21%
2000	0.038	$e^{76} \approx 1.0 \times 10^{33}$	41	56	37%

$M_{\min} \exp(N\Sigma_0)$ Growth, . N , .

8 SCX

8.1

56, SCX. Q_{ab} awaiting, $C_{\alpha\beta}$

D 24. $C = \sum_{k=1}^r \lambda_k v_k v_k^\top$, $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_r > 0$. k $v_k \in \mathbb{R}^M$, $v_{k,\alpha} \propto M$:

$$\sigma(x) = \sum_{k=1}^r c_k(x) \cdot v_k + \eta(x) \quad (44)$$

$c_k(x) \in \mathbb{R}^k$. . , $\eta(x)$.

Proposition 25. x :

$$d_{\text{eff}}(x) = \sum_{k=1}^r \mathbb{1}_{|c_k(x)| > \tau} \quad (45)$$

$\tau \cdot d_{\text{eff}}$.

8.2 Parisi

I RS: $\lambda_1 \gg \lambda_2 \approx 0$. — .

II 1RSB: $K \dots \lambda_1, \dots, \lambda_K$, . : , .

III fRSB: Gap.. . . .

8.3

Algorithm 1 SCX

Require: $M \{\mathcal{E}_\alpha\}_{\alpha=1}^M$, Task $\mathcal{T}$, δ

Ensure: Conclusion

```

1:  $C_{\alpha\beta} \leftarrow \mathbb{E}_{x \in \mathcal{T}}[\sigma_\alpha(x)\sigma_\beta(x)]$ 
2:  $\{(\lambda_k, v_k)\}_{k=1}^r \leftarrow \text{eig}(C)$ 
3: Parisi :
4: if  $\frac{\lambda_1}{\sum_k \lambda_k} > 1 - \delta$  then
5:   RS:
6: else if  $\exists K$  s.t.  $\frac{\sum_{k=1}^K \lambda_k}{\sum_k \lambda_k} > 1 - \delta$   $\frac{\lambda_K}{\lambda_{K+1}} \gg 1$  then
7:   1RSB:  $K, \rightarrow$ 
8: else
9:   fRSB: "...";
10: end if
11:  $R \leftarrow 1 - \exp(-M \cdot I(m_{\text{obs}}))$ 
12: return Conclusion,  $R$ ,

```

9 :

9.1

L1 : $\Sigma_0(e_\beta) > 0$ $\exp(N\Sigma_0) = \text{poly}(N)$ — $\therefore$.

L2 : $\Sigma_0(e_\beta) > 0$ Growth — $\therefore$.

L3 fRSB: , , — $\therefore$ — "...".

9.2

fRSB , :

Proposition 26 (—). fRSB , $\mathcal{E} \mathcal{E}'$, $M - 1$. *Conclusion:*

$$\lim_{M \rightarrow \infty} \mathbb{E} m_M(\mathcal{E}) - m_M(\mathcal{E}') > 0 \quad (46)$$

9.3

$$\lim_{\text{width} \rightarrow \infty} \Sigma_0(e) = 0 \quad \text{for all } e > e_{\min} \quad (47)$$

RS , — . : . $N_{\text{eff}} \Sigma_0(e)$.

10 Conclusion

1. , Hamiltonian $H(\theta) = \mathcal{L}(\theta) + \log Z$.
2. M Hamiltonian M , "...".
3. SCX, $M \rightarrow \infty$. $T_c = JJ$.

4. $Q_{ab} C_{\alpha\beta} \dots$
5. Parisi: RS, 1RSB fRSB.
6. : $M_{\min} \propto \exp(N\Sigma_0(e_\beta))$ — Growth.

Acknowledgments

A Proof

A.1 ??Proof

$$m = \tanh(\beta J m + \beta h) \quad (48)$$

$h = 0$. $m \ll 1$ Taylor:

$$m = \beta J m - \frac{(\beta J m)^3}{3} + \mathcal{O}(m^5) \quad (49)$$

$$m(1 - \beta J) + \frac{(\beta J)^3 m^3}{3} = 0 \quad (50)$$

$1 - \beta J < 0$, $T < J$ $\beta > 1/J$:

$$m = \pm \sqrt{\frac{3(\beta J - 1)}{(\beta J)^3}} \approx \pm \sqrt{31 - \frac{T}{T_c}}, \quad T \rightarrow T_c^- \quad (51)$$

$\beta = \beta_c(1 + \varepsilon)$, $\varepsilon = (T_c - T)/T \ll 1$. $M, m_M = \frac{1}{M} \sum_\alpha \sigma_\alpha$ $\text{Var}(m_M) = \frac{1-m^2}{M}$. $m \sqrt{\text{Var}(m_M)} \gtrsim 1$:

$$\sqrt{31 - \frac{T}{T_c}} \gtrsim \sqrt{\frac{1}{M}} \implies M \gtrsim \frac{1}{3(1 - T/T_c)} \quad (52)$$

$M_{\min} \propto |T - T_c|^{-1}$, $\nu = 1$. □

A.2 ??awaitingProof

, NTK : $f(x; \theta) = \sum_{i=1}^N \theta_i \phi_i(x)$, $\phi_i(x)$. x :

$$\mathbb{P}(\sigma_\alpha(x) = \sigma_\beta(x)) = \mathbb{P}(\text{sgn}(\theta_\alpha \cdot \phi(x)) = \text{sgn}(\theta_\beta \cdot \phi(x))) \quad (53)$$

$\theta_\alpha, \theta_\beta, \phi(x)$.:

$$\mathbb{P}(\sigma_\alpha = \sigma_\beta) = 1 - \frac{1}{\pi} \arccos \frac{\theta_\alpha \cdot \theta_\beta}{\|\theta_\alpha\| \|\theta_\beta\|} \quad (54)$$

$$C_{\alpha\beta} = \mathbb{E}[\sigma_\alpha \sigma_\beta] = \frac{2}{\pi} \arcsin \frac{\theta_\alpha \cdot \theta_\beta}{\|\theta_\alpha\| \|\theta_\beta\|} \quad (55)$$

$\tilde{Q}_{\alpha\beta} = \frac{\theta_\alpha \cdot \theta_\beta}{\|\theta_\alpha\| \|\theta_\beta\|}$. $C_{\alpha\beta} = g(\tilde{Q}_{\alpha\beta})$, $g(z) = \frac{2}{\pi} \arcsin(z)$ $[-1, 1]$. $|\mathcal{X}| \rightarrow \infty$, C . g . $N \rightarrow \infty$,
 $\tilde{Q} Q_{ab} = \frac{1}{N} \theta_a \cdot \theta_b$ awaiting Random Matrix, $\text{spec}(C)$ $\text{spec}(Q)$. □

A.3 ??

. $\mathcal{N} = \exp(N\Sigma_0(e_\beta))$, $k = 1, \dots, \mathcal{N}$. $\alpha \propto p_k$. $I_k^{(M)} = \mathbb{P}\{I_k^{(M)} = 1\} = (1 - p_k)^M$.
 k :

$$\mathbb{E} \sum_{k=1}^{\mathcal{N}} I_k^{(M)} = \sum_{k=1}^{\mathcal{N}} (1 - p_k)^M \quad (56)$$

p_k awaiting, $p_k = 1/\mathcal{N}$:

$$\mathbb{E}[] = \mathcal{N} \cdot 1 - \frac{1}{\mathcal{N}}^M \approx \mathcal{N} \cdot \exp -\frac{M}{\mathcal{N}} \quad (57)$$

$\leq \delta$:

$$\mathcal{N} e^{-M/\mathcal{N}} \leq \delta \implies M \geq \mathcal{N} \log \frac{\mathcal{N}}{\delta} \quad (58)$$

$\mathcal{N} \gg 1$, $\mathcal{N} = \exp(N\Sigma_0)$, . p_k , Bennett awaiting. $p_{(1)} \leq p_{(2)} \leq \dots \leq p_{(\mathcal{N})} \therefore$

$$\frac{\log \mathcal{N} - \log \delta}{-\log(1 - p_{(1)})} \leq M_{\min} \leq \frac{\log \mathcal{N} - \log \delta}{-\log(1 - p_{(\mathcal{N})})} \quad (59)$$

$p_{(k)} \propto k^{-\alpha} \alpha > 0$, Hessian, , .

□

B

Table 2:

$H(\theta)$	Hamiltonian
$\mathcal{L}(\theta)$	
$Z(\beta)$	
$\beta = 1/T$	
$F(\beta)$	
M	
n	
Q_{ab}	
$C_{\alpha\beta}$	
m_M	
q_{EA}	Edwards-Anderson
$q(x)$	Parisi
$\Sigma(e)$	
$\Sigma_0(e)$	
$\mathcal{N}_N(e)$	e
$\mathcal{H}_{ij}$	Hessian
$M_{\min}$	
d_{eff}	
RS	Replica Symmetric
RSB	Replica Symmetry Breaking
1RSB	
fRSB	

References

- [1] G. Parisi, “Infinite number of order parameters for spin-glasses,” *Phys. Rev. Lett.*, vol. 43, pp. 1754–1756, 1979.
- [2] G. Parisi, “A sequence of approximated solutions to the S-K model for spin glasses,” *J. Phys. A: Math. Gen.*, vol. 13, pp. L115–L121, 1980.
- [3] M. Mézard, G. Parisi, and M. A. Virasoro, *Spin Glass Theory and Beyond*. Singapore: World Scientific, 1987.
- [4] D. Sherrington and S. Kirkpatrick, “Solvable model of a spin-glass,” *Phys. Rev. Lett.*, vol. 35, pp. 1792–1796, 1975.
- [5] S. F. Edwards and P. W. Anderson, “Theory of spin glasses,” *J. Phys. F: Met. Phys.*, vol. 5, pp. 965–974, 1975.
- [6] A. Choromanska, M. Henaff, M. Mathieu, G. Ben Arous, and Y. LeCun, “The loss surfaces of multilayer networks,” in *Proc. AISTATS*, 2015, pp. 192–204.
- [7] Y. Dauphin, R. Pascanu, C. Gulcehre, K. Cho, S. Ganguli, and Y. Bengio, “Identifying and attacking the saddle point problem in high-dimensional non-convex optimization,” in *Proc. NeurIPS*, 2014, pp. 2933–2941.
- [8] V. Ros, G. Ben Arous, G. Biroli, and C. Cammarota, “Complex energy landscapes in spiked-tensor and simple glassy models: ruggedness, arrangements of local minima, and phase transitions,” *Phys. Rev. X*, vol. 9, p. 011003, 2019.
- [9] A. Engel and C. Van den Broeck, *Statistical Mechanics of Learning*. Cambridge: Cambridge University Press, 2001.
- [10] E. Levin and M. Z. Ziegler, “The phase transition in random committee machines,” *J. Phys. A: Math. Theor.*, vol. 52, p. 384001, 2019.
- [11] A. Jacot, F. Gabriel, and C. Hongler, “Neural tangent kernel: Convergence and generalization in neural networks,” in *Proc. NeurIPS*, 2018, pp. 8571–8580.
- [12] Y. V. Fyodorov, “Complexity of random energy landscapes, glass transition, and absolute value of the spectral determinant of random matrices,” *Phys. Rev. Lett.*, vol. 92, p. 240601, 2004.
- [13] G. Ben Arous, R. Gheissari, and A. Jagannath, “Bounding flows for spherical spin glass dynamics,” *Comm. Math. Phys.*, vol. 373, pp. 1011–1048, 2019.
- [14] M. Baity-Jesi, L. Sagun, M. Geiger, S. Spigler, G. Ben Arous, C. Cammarota, Y. LeCun, and M. Wyart, “Comparing dynamics: Deep neural networks versus glassy systems,” *J. Stat. Mech.*, vol. 2019, p. 124013, 2019.
- [15] P. Baldi and S. S. Sadowski, “A theory of local minima in deep learning,” in *Proc. ICLR Workshop*, 2016.
- [16] M. Geiger, S. Spigler, S. d’Ascoli, L. Sagun, M. Baity-Jesi, G. Biroli, and M. Wyart, “Jamming transition as a paradigm to understand the loss landscape of deep neural networks,” *Phys. Rev. E*, vol. 101, p. 012115, 2020.

- [17] L. Sagun, L. Bottou, and Y. LeCun, “Eigenvalues of the Hessian in deep learning: Singularity and beyond,” in *Proc. ICLR Workshop*, 2017.
- [18] T. Castellani and A. Cavagna, “Spin-glass theory for pedestrians,” *J. Stat. Mech.*, vol. 2005, p. P05012, 2005.
- [19] M. Talagrand, *Spin Glasses: A Challenge for Mathematicians*. Berlin: Springer, 2003.
- [20] F. Guerra, “Broken replica symmetry bounds in the mean field spin glass model,” *Comm. Math. Phys.*, vol. 233, pp. 1–12, 2003.
- [21] L. Sagun, U. Evci, V. U. Güney, Y. Dauphin, and L. Bottou, “Empirical analysis of the Hessian of over-parametrized neural networks,” in *Proc. ICLR Workshop*, 2018.
- [22] J. Pennington and Y. Bahri, “Geometry of neural network loss surfaces via random matrix theory,” in *Proc. ICML*, 2017, pp. 2798–2806.
- [23] S. Franz, G. Parisi, M. Sevelev, P. Urbani, and F. Zamponi, “Universality of the SAT-UNSAT (jamming) threshold in non-convex continuous constraint satisfaction problems,” *SciPost Phys.*, vol. 2, p. 019, 2017.
- [24] Y. LeCun, Y. Bengio, and G. Hinton, “Deep learning,” *Nature*, vol. 521, pp. 436–444, 2015.
- [25] S. Amari, “Any target function exists in a neighborhood of any sufficiently wide random network: A geometrical perspective,” *Neural Computation*, vol. 32, pp. 1431–1447, 2020.